

APM1220: Applied Multivariate Data Analysis
Formative Assessment 1 (FA 1)

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Problem 2.2

Given the matrices:

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$$

- (a) **Find 5A:** Multiply each entry of matrix A by the scalar 5:

$$5A = 5 \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 5(-1) & 5(3) \\ 5(4) & 5(2) \end{bmatrix} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

- (b) **Find BA:** Matrix B is dimension 3×2 and matrix A is 2×2 . The inner dimensions match ($2 = 2$), so the product is defined and yields a 3×2 matrix:

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} (4)(-1) + (-3)(4) & (4)(3) + (-3)(2) \\ (1)(-1) + (-2)(4) & (1)(3) + (-2)(2) \\ (-2)(-1) + (0)(4) & (-2)(3) + (0)(2) \end{bmatrix}$$

$$BA = \begin{bmatrix} -4 - 12 & 12 - 6 \\ -1 - 8 & 3 - 4 \\ 2 + 0 & -6 + 0 \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

- (c) **Find A'B':** First, find the transposes. Matrix A' is 2×2 and B' is 2×3 . The inner dimensions match ($2 = 2$), yielding a 2×3 matrix:

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}, \quad B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$$A'B' = \begin{bmatrix} (-1)(4) + (4)(-3) & (-1)(1) + (4)(-2) & (-1)(-2) + (4)(0) \\ (3)(4) + (2)(-3) & (3)(1) + (2)(-2) & (3)(-2) + (2)(0) \end{bmatrix}$$

$$A'B' = \begin{bmatrix} -4 - 12 & -1 - 8 & 2 + 0 \\ 12 - 6 & 3 - 4 & -6 + 0 \end{bmatrix} = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

- (d) **Find C'B:** Matrix C' is 1×3 and B is 3×2 . Inner dimensions match ($3 = 3$), yielding a 1×2 matrix:

$$C'B = [5 \quad -4 \quad 2] \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} = [(5)(4) + (-4)(1) + (2)(-2) \quad (5)(-3) + (-4)(-2) + (2)(0)]$$

$$C'B = [20 - 4 - 4 \quad -15 + 8 + 0] = [12 \quad -7]$$

- (e) **Is AB defined?** Matrix A is 2×2 and matrix B is 3×2 . Because the inner dimensions do not match ($2 \neq 3$), **AB is not defined.**

Problem 2.4

When A^{-1} and B^{-1} exist, prove each of the following:

- (a) **Prove that** $(A')^{-1} = (A^{-1})'$

Proof: By the definition of an invertible matrix, A satisfies $AA^{-1} = I$ and $A^{-1}A = I$. Taking the transpose of both sides of $AA^{-1} = I$ and noting that the transpose of the identity matrix is itself ($I' = I$):

$$(AA^{-1})' = I' \implies (A^{-1})'A' = I$$

Similarly, taking the transpose of the second relation:

$$(A^{-1}A)' = I' \implies A'(A^{-1})' = I$$

Since both products equal I , by the uniqueness of the matrix inverse, we conclude that the inverse of A' must be $(A^{-1})'$. Therefore, $(A')^{-1} = (A^{-1})'$. $\square$

- (b) **Prove that** $(AB)^{-1} = B^{-1}A^{-1}$

Proof: To prove this, we must show that multiplying AB by $B^{-1}A^{-1}$ yields the identity matrix I . Consider the product $(B^{-1}A^{-1})(AB)$. Using the associative property of matrix multiplication:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$$

Since $A^{-1}A = I$, we can substitute I :

$$B^{-1}(I)B = B^{-1}B$$

Since $B^{-1}B = I$, we have shown that $(B^{-1}A^{-1})(AB) = I$. Checking the reverse multiplication:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Thus, by definition of the inverse, $(AB)^{-1} = B^{-1}A^{-1}$. $\square$

Problem 2.8

Given the matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

- (1) **Find Eigenvalues:** Solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{bmatrix} = 0$$

$$(1 - \lambda)(-2 - \lambda) - (2)(2) = 0$$

$$-2 - \lambda + 2\lambda + \lambda^2 - 4 = 0 \implies \lambda^2 + \lambda - 6 = 0$$

Factoring the quadratic yields $(\lambda + 3)(\lambda - 2) = 0$. The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -3$.

- (2) **Find Associated Normalized Eigenvectors:**

- For $\lambda_1 = 2$:

$$(A - 2I)x = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $-x_1 + 2x_2 = 0 \implies x_1 = 2x_2$. Let $x_2 = 1, x_1 = 2$. The unnormalized vector is $[2, 1]'$. Length $= \sqrt{2^2 + 1^2} = \sqrt{5}$. Normalized eigenvector: $\mathbf{e}_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$

- For $\lambda_2 = -3$:

$$(A - (-3)I)x = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $2x_1 + x_2 = 0 \implies x_2 = -2x_1$. Let $x_1 = 1, x_2 = -2$. The unnormalized vector is $[1, -2]'$. Length $= \sqrt{1^2 + (-2)^2} = \sqrt{5}$. Normalized eigenvector: $\mathbf{e}_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix}$

- (3) **Determine Spectral Decomposition:** Using $A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$:

$$A = 2 \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} + (-3) \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix}$$

$$A = 2 \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix} - 3 \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} \frac{8}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} \end{bmatrix} + \begin{bmatrix} -\frac{3}{5} & \frac{6}{5} \\ \frac{6}{5} & -\frac{12}{5} \end{bmatrix}$$

Problem 2.11

Show that the determinant of a $p \times p$ diagonal matrix $A = \{a_{ij}\}$ with $a_{ij} = 0$ for $i \neq j$, is given by $|A| = a_{11}a_{22} \cdots a_{pp}$.

Proof: By the definition of cofactor expansion (Definition 2A.24) along the first row, we evaluate the determinant. Since all non-diagonal elements in the first row are zero except a_{11} , the expansion simplifies to:

$$|A| = a_{11}A_{11} + 0 + \cdots + 0 = a_{11}A_{11}$$

where A_{11} is the determinant of the $(p-1) \times (p-1)$ submatrix obtained by deleting the first row and first column of A . Because A is a diagonal matrix, the submatrix whose determinant is A_{11} is also a diagonal matrix, consisting of the diagonal elements $a_{22}, a_{33}, \dots, a_{pp}$. Repeating this cofactor expansion process successively for each submatrix down to the final 1×1 trailing submatrix yields:

$$|A| = a_{11} \cdot a_{22} \cdot a_{33} \cdots a_{pp} = \prod_{i=1}^p a_{ii}$$

□

Problem 2.24

Given covariance matrix $\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$:

- (a) **Find Σ^{-1}** : Because Σ is a diagonal matrix, its inverse is the diagonal matrix of the reciprocals of the original diagonal entries:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (b) **Eigenvalues and eigenvectors of Σ** : For a diagonal matrix, the eigenvalues are the diagonal entries, and the standard basis vectors are the corresponding eigenvectors:

$$\lambda_1 = 4, \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \lambda_2 = 9, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3 = 1, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

- (c) **Eigenvalues and eigenvectors of Σ^{-1}** : Similarly, for the diagonal inverse matrix:

$$\lambda_1 = \frac{1}{4}, \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \lambda_2 = \frac{1}{9}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3 = 1, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 2.30

Given partitioned vectors and matrices: $\mu_X = [4, 3, 2, 1]'$ partitioned into $X^{(1)}$ and $X^{(2)}$, meaning $\mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Σ_X is partitioned into $\Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $\Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$, and $\Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$. $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$.

(a) $\mathbf{E}(\mathbf{X}^{(1)}) = \mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(b) $\mathbf{E}(\mathbf{A}\mathbf{X}^{(1)}) = \mathbf{A}\mu^{(1)} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \mathbf{1}(4) + \mathbf{2}(3) = \mathbf{10}$

(c) $\mathbf{Cov}(\mathbf{X}^{(1)}) = \Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

(d) **Find** $Cov(AX^{(1)})$: $Cov(AX^{(1)}) = A\Sigma_{11}A'$

$$= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = [3(1) + 2(2)] = [7]$$

(e) $\mathbf{E}(\mathbf{X}^{(2)}) = \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(f) **Find** $E(BX^{(2)})$: $E(BX^{(2)}) = B\mu^{(2)} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(2) - 2(1) \\ 2(2) - 1(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

(g) $\mathbf{Cov}(\mathbf{X}^{(2)}) = \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$

(h) **Find** $Cov(BX^{(2)})$: $Cov(BX^{(2)}) = B\Sigma_{22}B' = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$$= \begin{bmatrix} (1)(9) + (-2)(-2) & (1)(-2) + (-2)(4) \\ (2)(9) + (-1)(-2) & (2)(-2) + (-1)(4) \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 13(1) + (-10)(-2) & 13(2) + (-10)(-1) \\ 20(1) + (-8)(-2) & 20(2) + (-8)(-1) \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

(i) $\mathbf{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$

(j) **Find** $Cov(AX^{(1)}, BX^{(2)})$: $Cov(AX^{(1)}, BX^{(2)}) = A\Sigma_{12}B' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 1(2) + 2(1) & 1(2) + 2(0) \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4(1) + 2(-2) & 4(2) + 2(-1) \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}$$

Problem 2.41

Given $\mu_X = [3, 2, -2, 0]'$, $\Sigma_X = 3I_4$, and transformation matrix $A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}$:

(a) **Find** $E(AX)$:

$$E(AX) = A\mu_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1(3) - 1(2) + 0 + 0 \\ 1(3) + 1(2) - 2(-2) + 0 \\ 1(3) + 1(2) + 1(-2) - 3(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

- (b) **Find $Cov(AX)$:** $Cov(AX) = A\Sigma_X A' = A(3I_4)A' = 3AA'$ First, find AA' (Dimensions: 3×4 multiplied by 4×3 yields 3×3):

$$\begin{aligned}
 AA' &= \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix} \\
 &= \begin{bmatrix} (1)(1) + (-1)(-1) & (1)(1) + (-1)(1) & (1)(1) + (-1)(1) \\ (1)(1) + (1)(-1) & (1)(1) + (1)(1) + (-2)(-2) & (1)(1) + (1)(1) + (-2)(1) \\ (1)(1) + (1)(-1) & (1)(1) + (1)(1) + (1)(-2) & (1)(1) + (1)(1) + (1)(1) + (-3)(-3) \end{bmatrix} \\
 &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}
 \end{aligned}$$

Multiply by scalar 3:

$$Cov(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

- (c) **Which pairs of linear combinations have zero covariances?** Since $Cov(AX)$ is a diagonal matrix, all off-diagonal elements are 0. Therefore, **all pairs of distinct linear combinations** have zero covariances (they are mutually uncorrelated).